

FROM PHYSICAL LAWS TO SIMULATIONS

Finite Element Methods: Foundations

From weak forms to error estimates

Weak forms $\longrightarrow$ finite elements $\longrightarrow$ reliable solutions

Why FEM?

Real models are rarely simple

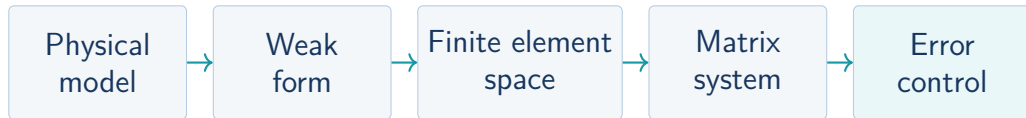
- ▶ Complicated geometry and material variation.
- ▶ Fields such as temperature, displacement, and pressure.

FEM makes them computable

- ▶ Local polynomial approximations on a mesh.
- ▶ Sparse systems with a route to error control.

Along with the solution, we need an estimate of its error.

From a PDE to a simulation



One model problem will connect every step.

Our model problem

Poisson equation

$$-\Delta u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega.$$

- ▶ u : the unknown field.
- ▶ f : the prescribed source.

Why this example?

We can derive its weak form, assemble its stiffness matrix, and estimate the error.

A warm-up in one dimension

Start with the differential equation

$$-u'' = f \quad \text{on } (0, 1), \quad u(0) = u(1) = 0.$$

- ▶ Multiply by a test function v with $v(0) = v(1) = 0$.
- ▶ Integrate over the interval.

$$\int_0^1 (-u'')v \, dx = \int_0^1 fv \, dx.$$

Integration by parts

$$\int_0^1 (-u'')v \, dx = \int_0^1 u'v' \, dx - [u'v]_0^1.$$

The test function vanishes at both ends

$$\int_0^1 u'v' \, dx = \int_0^1 fv \, dx.$$

Integration by parts leaves only first derivatives of u and v .

The weak Poisson problem

Find $u \in V = H_0^1(\Omega)$ such that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in V.$$

Energy form

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx.$$

Load functional

$$\ell(v) = \int_{\Omega} f v \, dx.$$

We write this as $a(u, v) = \ell(v)$.

What belongs to the function space?

$$H_0^1(\Omega)$$

- ▶ The function and its first weak derivatives are square integrable.
- ▶ Its boundary value is zero in the trace sense.

Why this matters for FEM

Continuous piecewise-linear functions can belong to this space even though their gradients jump between cells.

Continuity and coercivity

Continuity: bounded response

$$|a(w, v)| \leq M \|w\|_V \|v\|_V.$$

The constant M bounds the size of $a(w, v)$.

Coercivity: energy controls the norm

$$a(v, v) \geq \alpha \|v\|_V^2, \quad \alpha > 0.$$

The energy controls the size of the function.

Coercivity plays the role of positive definiteness.

Lax–Milgram: the result we use

- ▶ V is a Hilbert space.
- ▶ a is continuous and coercive.
- ▶ ℓ is a continuous linear functional.

Existence, uniqueness, and stability

There is a unique $u \in V$ with $a(u, v) = \ell(v)$ for every $v \in V$, and

$$\|u\|_V \leq \alpha^{-1} \|\ell\|_{V'}.$$

Poisson satisfies the assumptions

Use the energy norm $\|v\|_V = \|\nabla v\|_{L^2}$ on $H_0^1(\Omega)$.

$$|a(w, v)| \leq \|w\|_V \|v\|_V, \quad M = 1,$$

$$a(v, v) = \|v\|_V^2, \quad \alpha = 1.$$

The load is bounded

For $f \in L^2(\Omega)$, Poincaré's inequality gives

$$|\ell(v)| \leq C_P \|f\|_{L^2} \|v\|_V.$$

Mixed boundary data: where does the flux go?

A one-dimensional example

$$-u'' = f, \quad u(0) = 0, \quad u'(1) = g.$$

Choose test functions with $v(0) = 0$, but do not require $v(1) = 0$.

$$\int_0^1 u' v' \, dx = \int_0^1 f v \, dx + g v(1).$$

Neumann data enter the load through the boundary term.

A simple well-posedness check

For reaction–diffusion on $V = H^1(0, 1)$, take

$$a(u, v) = \int_0^1 u' v' dx + \int_0^1 uv dx.$$

Immediate energy identity

$$a(u, u) = \|u\|_{H^1}^2 \quad \implies \quad M = \alpha = 1.$$

What changes without the mass term?

On all of H^1 , constant functions have zero gradient. Boundary conditions or another constraint are then needed for coercivity.

From infinite to finite dimension

Continuous problem

Find $u \in V$ with

$$a(u, v) = \ell(v) \quad \forall v \in V.$$

Galerkin problem

Choose $V_h \subset V$ and find $u_h \in V_h$ with

$$a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in V_h.$$

Keep the weak form. Restrict the trial and test space.

Stability passes to the discrete problem

Because $V_h \subset V$

- ▶ The same continuity and coercivity bounds still hold.
- ▶ The discrete problem therefore has a unique solution.

$$\|u_h\|_V \leq \alpha^{-1} \|\ell\|_{V'}.$$

The discrete problem is stable. Its accuracy depends on the choice of V_h .

Galerkin orthogonality

Test both equations with the same $v_h \in V_h$ and subtract:

$$a(u, v_h) - a(u_h, v_h) = 0.$$

The key identity

$$a(u - u_h, v_h) = 0 \quad \forall v_h \in V_h.$$

For the symmetric Poisson form, the error is orthogonal to V_h in the energy inner product.

Céa's lemma: nearly the best approximation

For a continuous, coercive form and a conforming space $V_h \subset V$,

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V.$$

Interpretation

The FEM error is at most M/α times the best error in V_h .

To improve accuracy, improve the approximation space.

For Poisson: best approximation in energy

The symmetric form defines $\|v\|_a^2 = a(v, v)$.

$$\|u - u_h\|_a = \min_{v_h \in V_h} \|u - v_h\|_a.$$

An energy projection

Among all functions in V_h , the Galerkin solution is closest to u in the energy norm.

A first mesh in one dimension

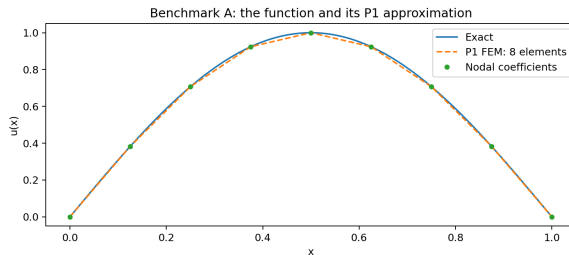
Split $[0, 1]$ into elements

$$K_i = [x_{i-1}, x_i].$$

$$h = \max_i |K_i|.$$

For P_1

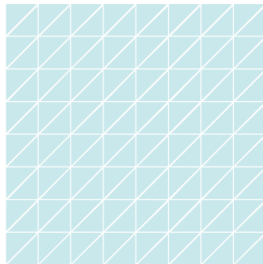
The approximation is linear on each interval and determined by nodal values.



A triangular mesh

- ▶ The cells form a mesh $\mathcal{T}_h$.
- ▶ Neighboring cells share a full edge or a vertex.
- ▶ h measures the largest cell size.

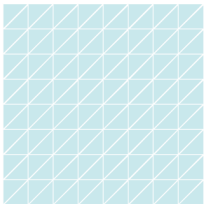
(a)
structured, one diagonal



128 cells · 81 vertices
 $\max h_K/\rho_K = 2.4$

Mesh quality is not just mesh size

(a) structured, one diagonal



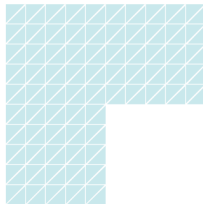
128 cells · 81 vertices
 $\max h_K/\rho_K = 2.4$

(b) unstructured, Lloyd relaxed



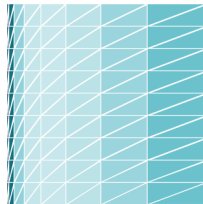
394 cells · 226 vertices
 $\max h_K/\rho_K = 3.1$

(c) L-shape: re-entrant corner



150 cells · 96 vertices
 $\max h_K/\rho_K = 2.4$

(d) graded in x only



162 cells · 100 vertices
 $\max h_K/\rho_K = 81.5$

shape regularity lost



Standard estimates assume shape-regular meshes: avoid arbitrarily thin cells.

What does P_k mean?

$P_k(K)$ contains polynomials of total degree at most k on one cell.

P_1 on a triangle

$$p(x, y) = a + bx + cy.$$

Three coefficients.

P_2 on a triangle

$$p(x, y) = a + bx + cy + dx^2 + exy + fy^2.$$

Six coefficients.

Raising the degree gives more freedom to approximate u on the same cell.

Lagrange elements: the nodal idea

$$N_i(\xi_j) = \delta_{ij}.$$

One node, one basis function

A basis function is one at its own node and zero at the other nodes.

Nodal values become coefficients

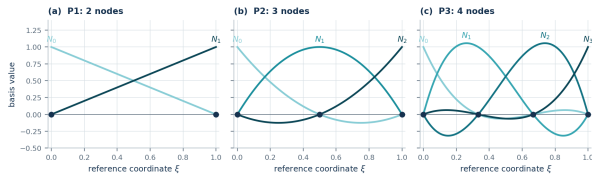
$$p(\xi) = \sum_i p(\xi_i) N_i(\xi).$$

The nodes determine the polynomial.

Two shape functions for a P_1 interval

$$\hat{\phi}_0(\xi) = 1 - \xi, \quad \hat{\phi}_1(\xi) = \xi.$$

Two endpoint values determine one line.

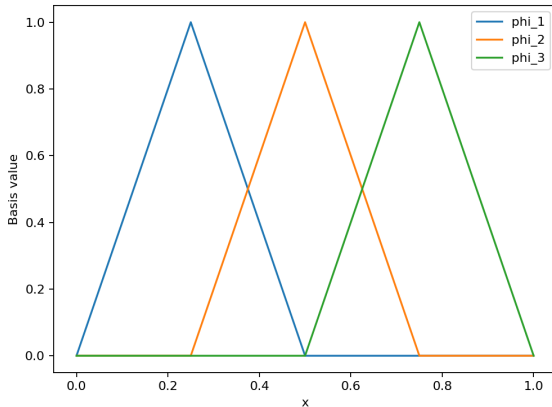


Global P_1 : hat functions

$$\phi_i(x_j) = \delta_{ij}.$$

Each hat is supported on the two elements next to its node.

$$u_h = \sum_i U_i \phi_i, \quad U_i = u_h(x_i).$$



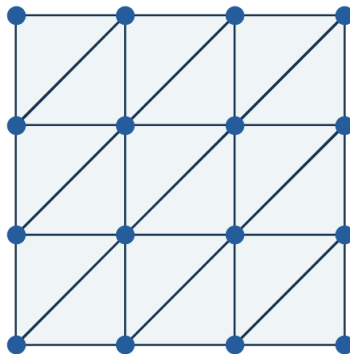
P_1 on a triangle

Three vertex values

A linear function $p(x, y) = a + bx + cy$ has three coefficients.

Each local shape function is one at one vertex and zero at the other two.

(a) P1: 16 dofs



3 per cell · vertices

How local pieces form a conforming space

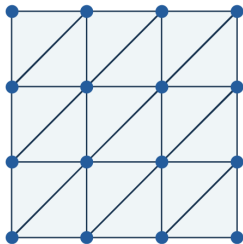
- ▶ Use a polynomial on each cell.
- ▶ Share nodal values across cell interfaces.
- ▶ Set the boundary nodal values to zero.

$$V_h \subset H_0^1(\Omega).$$

A conforming space lies inside the energy space, including its boundary conditions.

P_1 versus P_2 : more information on each cell

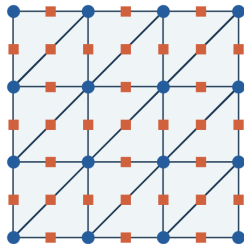
(a) P_1 : 16 dofs



3 per cell · vertices

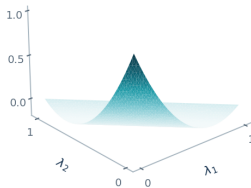
● vertex dof ■ edge dof

(b) P_2 : 49 dofs

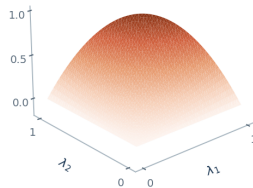


6 per cell · vertices + edge midpoints

(c) vertex: $\lambda_0(2\lambda_0 - 1)$



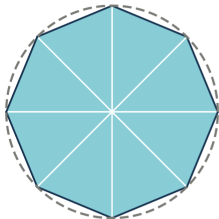
(d) edge: $4\lambda_1\lambda_2$



Higher degree can improve accuracy for smooth solutions, but adds unknowns and cost.

Reference elements: reuse the same local construction

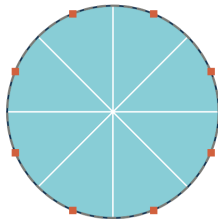
(a) 8 straight-sided cells



worst geometry gap $7.61\text{e-}02$

— true boundary

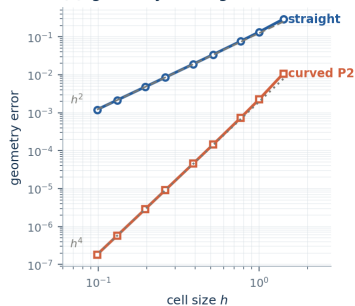
(b) 8 isoparametric P2 cells



worst geometry gap $7.25\text{e-}04$

■ edge node on the true boundary

(c) geometry error against cell size



Define shape functions once on $\hat{K}$.

Map them to each physical cell using F_K .

From the weak form to a matrix system

Expand the unknown and test with each basis function:

$$u_h = \sum_{j=1}^N U_j \phi_j \quad \Longrightarrow \quad \sum_{j=1}^N a(\phi_j, \phi_i) U_j = \ell(\phi_i).$$

The linear system

$$\boxed{AU = F}, \quad A_{ij} = a(\phi_j, \phi_i), \quad F_i = \ell(\phi_i).$$

For Poisson, A is the stiffness matrix.

One element stiffness matrix

On a P_1 interval of length h_K , the basis derivatives are $-1/h_K$ and $1/h_K$.

$$A^K = \frac{1}{h_K} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

Every entry is a local integral of two basis-function derivatives.

One element load vector

For a constant source $f = 1$ on an interval of length h_K ,

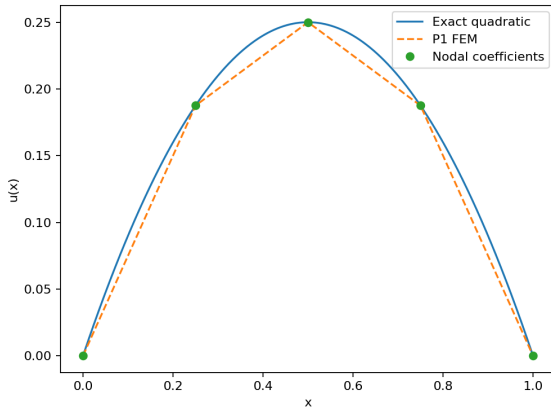
$$F_m^K = \int_K \phi_m dx \quad \Rightarrow \quad \boxed{F^K = \frac{h_K}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} .}$$

Geometric interpretation

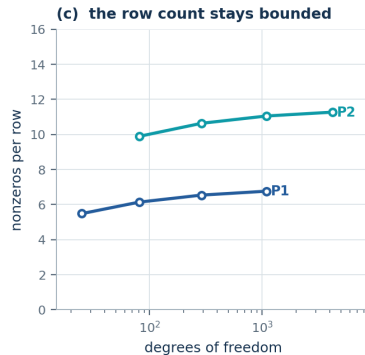
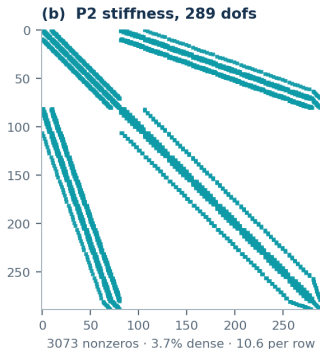
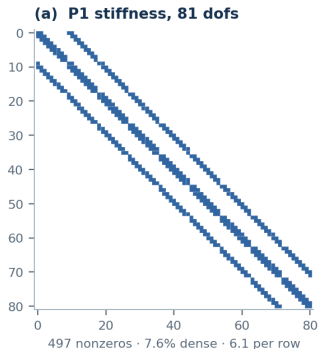
Each basis function is a triangle of base h_K and height one, so its integral is $h_K/2$.

Local-to-global assembly

- ▶ Compute a small matrix and load vector on each cell.
- ▶ Map local node numbers to global unknowns.
- ▶ Add contributions at shared unknowns.



Why the global matrix is sparse



Only nearby basis functions interact

If ϕ_i and ϕ_j have no common cell of support, then $A_{ij} = 0$.

Refinement enlarges the system without making every unknown interact with every other one.

Boundary conditions in the discrete system

Dirichlet data

- ▶ Fix the boundary nodal values.
- ▶ For nonzero data, adjust the interior right-hand side.

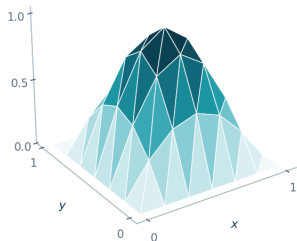
Neumann data

- ▶ Keep the boundary flux term in the load.
- ▶ Do not prescribe the corresponding field value.

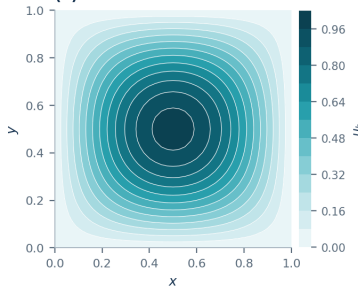
Impose the boundary conditions consistently with the weak form.

What a two-dimensional P_1 field looks like

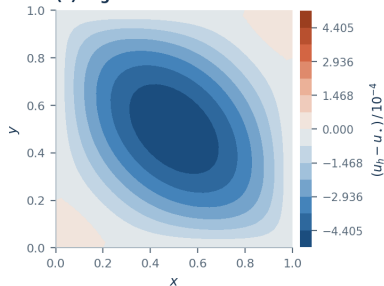
(a) P_1 on 72 cells: a faceted surface



(b) P_1 on 3200 cells



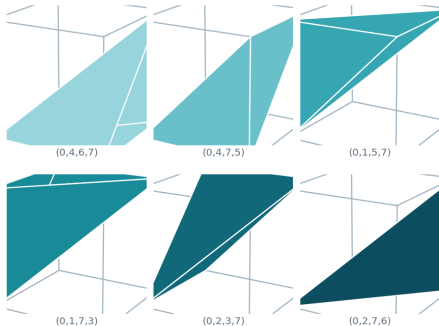
(c) signed nodal error



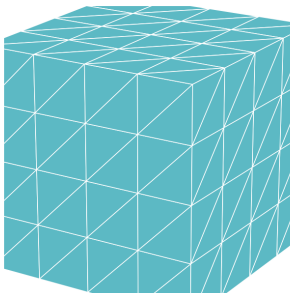
The field is continuous and planar on each triangle. Refinement makes the facets smaller.

The same construction extends to three dimensions

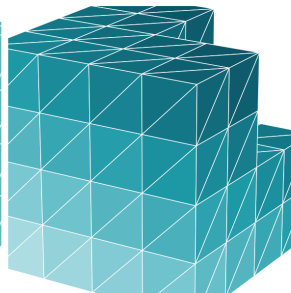
(a) one cube = six tetrahedra, all sharing the diagonal 0-7



(b) 384 cells, 125 vertices



(c) one octant removed: 336 cells



Tetrahedra replace triangles.

The weak form and local assembly follow the same pattern.

Refinement introduces unknowns much faster in 3D.

From best approximation to a convergence rate

The missing step

Céa bounds FEM error by the best approximation error. We still need to know how that approximation improves as h decreases.

$$\underbrace{\text{Stability}}_{\text{Galerkin / Céa}} + \underbrace{\text{Approximation}}_{\text{Interpolation}} \implies \text{Convergence rate.}$$

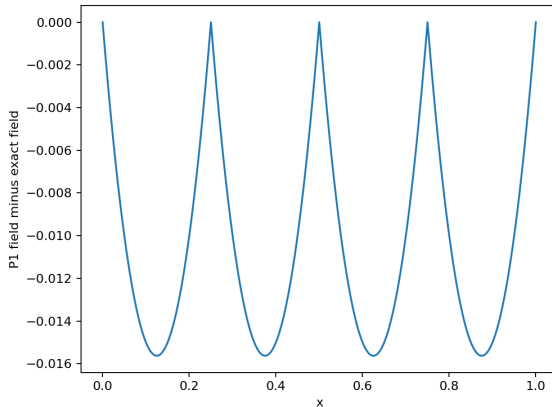
The convergence rate depends on mesh size, polynomial degree, and solution smoothness.

Interpolation: exact at nodes, not between them

For smooth u , the nodal interpolant $I_h u$ matches its values at the finite element nodes.

Interpolant and computed solution

$I_h u$ is an approximation tool. The computed FEM solution u_h is defined by the Galerkin equations.



The basic P_1 interpolation estimates

For $u \in H^2(\Omega)$ on a shape-regular mesh,

$$\|u - I_h u\|_{H^1} \leq Ch |u|_{H^2},$$

$$\|u - I_h u\|_{L^2} \leq Ch^2 |u|_{H^2}.$$

The constant is independent of mesh size. The two norms measure different errors.

Higher degree improves the approximation order

For sufficiently smooth u and degree- k elements,

$$\|u - I_h u\|_{H^1} \lesssim h^k |u|_{H^{k+1}},$$

$$\|u - I_h u\|_{L^2} \lesssim h^{k+1} |u|_{H^{k+1}}.$$

The higher rate requires derivatives of u through order $k + 1$.

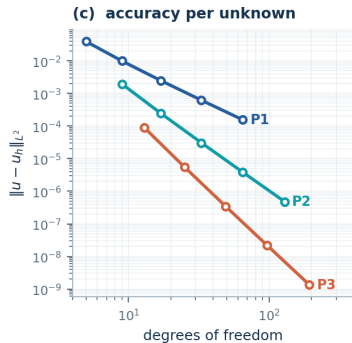
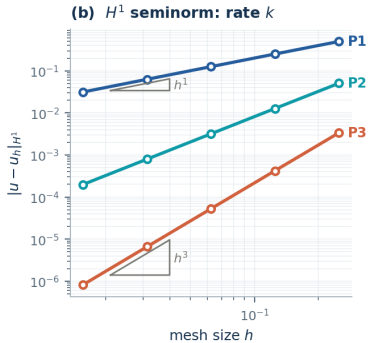
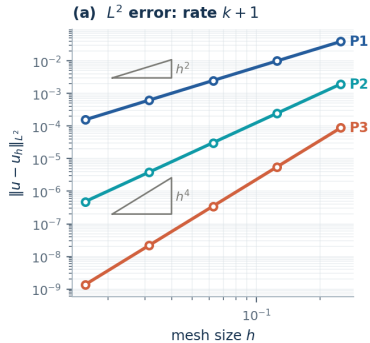
Energy-error rates for the FEM solution

Combine Céa's lemma with interpolation:

$$\|u - u_h\|_{H^1} \leq C \|u - I_h u\|_{H^1}.$$

Element	Smoothness needed	Energy / H^1 error
P_1	$u \in H^2$	$O(h)$
P_2	$u \in H^3$	$O(h^2)$

How to read a convergence plot



If $E(h) \approx Ch^p$, halving h reduces the error by about 2^p .

Regularity can limit the rate

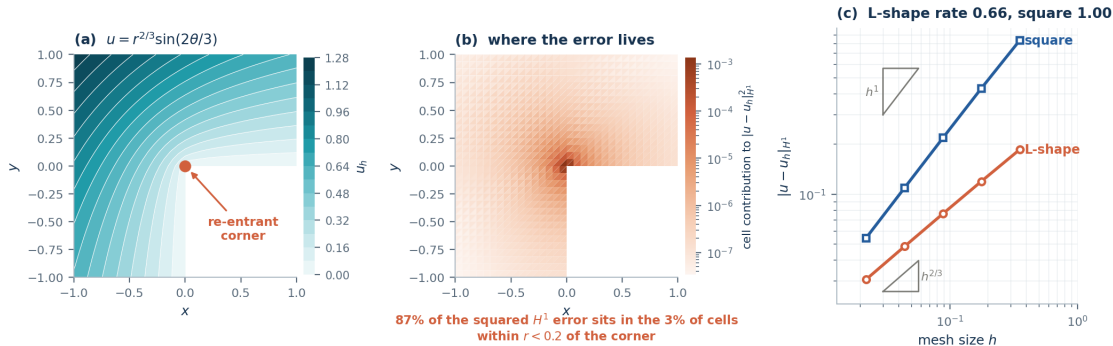
The nominal rate has a condition

The degree- k estimate uses derivatives of u up to order $k + 1$. Without them, the expected rate may not appear.

- ▶ Re-entrant corners and changing boundary conditions.
- ▶ Material interfaces, rough coefficients, or singular sources.

Increasing polynomial degree alone does not remove a singularity.

An L-shaped domain: the re-entrant corner



A re-entrant corner

The solution may belong to H^1 but not H^2 . The standard first-order P_1 energy estimate then loses its smoothness assumption.

Local refinement targets the small region that dominates the error.

Why the L^2 error can converge faster

Aubin–Nitsche: the useful conclusion

With sufficient primal and dual regularity,

$$\|u - u_h\|_{L^2} \leq Ch \|u - u_h\|_{H^1}.$$

For smooth degree- k FEM, this gives $O(h^{k+1})$ in L^2 .

The extra order is conditional

Non-convex domains or rough coefficients can reduce the gain.

Two kinds of error information

A priori: predict the trend

- ▶ Uses h , degree, and solution regularity.
- ▶ Explains asymptotic convergence rates.

A posteriori: assess this computation

- ▶ Uses the computed u_h and problem data.
- ▶ Locates where refinement is useful.

The exact solution is unknown in practice; computable indicators help guide the mesh.

Element residuals: the local equation mismatch

The exact solution satisfies $-\Delta u = f$. Inside each element, define

$$R_K = f + \Delta u_h.$$

For affine P_1 elements

$\Delta u_h = 0$ inside each cell, so $R_K = f$ for the basic Poisson problem. Edge information is also needed.

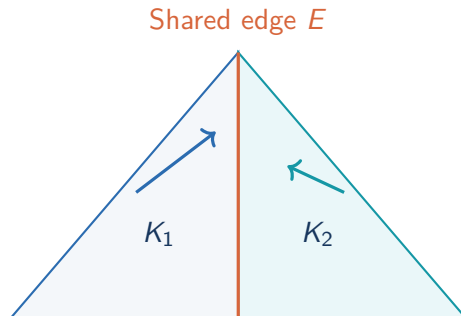
Flux jumps add the missing interface information

$$J_E = [\nabla u_h \cdot n]_E.$$

Although u_h is continuous, its gradient usually changes across cell boundaries.

Local indicator

Large flux jumps can signal insufficient resolution.



The field is continuous;
its gradient need not be.

What a useful error estimator measures

Combine cell and interface information

Each local indicator η_K combines mesh-weighted residual and flux-jump contributions.

$$\eta = \left(\sum_K \eta_K^2 \right)^{1/2}.$$

Reliability

The estimator controls the true error, up to a constant.

Efficiency

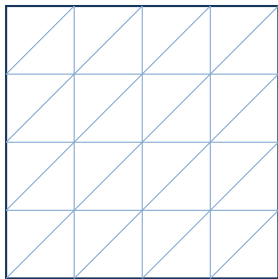
It also gives a lower bound for the error, up to constants and data-approximation terms.

The adaptive FEM loop

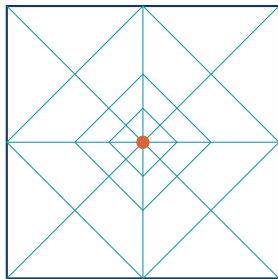


Add unknowns where the estimated error is largest.

Uniform versus adaptive refinement



Uniform resolution



Local refinement (schematic)

Uniform refinement adds resolution everywhere. Adaptivity concentrates it near difficult features.

Marking elements: a small example

Select cells whose squared indicators account for a chosen fraction of the total.

$$\eta_K^2 : \quad 16, 9, 4, 1 \quad \implies \quad \text{total} = 30.$$

Target: 50% of the estimated squared error

The target is 15. The largest contribution is 16, so that element alone reaches the target.

This illustrates bulk marking, also called Dörfler marking.

From Poisson to the heat equation

$$\begin{aligned} u_t - \Delta u &= f \quad \text{in } \Omega \times (0, T), \\ u &= 0 \quad \text{on } \partial\Omega, \quad u(\cdot, 0) = u_0. \end{aligned}$$

Two discretization choices

Approximate the field in space, then advance its coefficients in time.

The weak heat equation

Test and integrate in space, just as for Poisson:

$$(u_t, v)_{L^2} + a(u, v) = (f, v)_{L^2} \quad \forall v \in H_0^1(\Omega).$$

What is new?

The time derivative appears in an L^2 inner product. Time is still continuous.

Discretize only in space

Choose $V_h \subset H_0^1(\Omega)$ and seek $u_h(t) \in V_h$:

$$(\partial_t u_h, v_h)_{L^2} + a(u_h, v_h) = (f, v_h)_{L^2} \quad \forall v_h \in V_h.$$

Semidiscrete approximation

The basis functions are fixed in space. Their coefficients $U_j(t)$ vary continuously in time.

Space discretization produces an ODE system

$$u_h(t) = \sum_j U_j(t) \phi_j \quad \Rightarrow \quad \boxed{M\dot{U} + AU = F.}$$

Mass matrix

$$M_{ij} = (\phi_j, \phi_i)_{L^2}.$$

Stiffness matrix

$$A_{ij} = a(\phi_j, \phi_i).$$

This is the method-of-lines viewpoint.

Time levels and initial data

Choose the time grid

$$t_n = n\Delta t, \quad U^n \approx U(t_n).$$

Represent the starting field

Choose $u_h^0 \in V_h$ by interpolation or projection of u_0 .

A time-stepping method advances the coefficient vector from one level to the next.

Forward Euler

Evaluate the stiffness term at the current time level:

$$M \frac{U^{n+1} - U^n}{\Delta t} + AU^n = F^n.$$

- ▶ First order in time.
- ▶ The update still requires handling the mass matrix M .

Its diffusion stability limit depends on the spatial mesh.

The explicit diffusion time-step restriction

$$\Delta t \lesssim Ch^2.$$

A practical consequence

On comparable quasi-uniform meshes, halving h requires roughly four times as many time steps over the same time interval.

Stability is not accuracy

A step that is stable can still be too large to resolve the time evolution.

Backward Euler

$$(M + \Delta t A)U^{n+1} = MU^n + \Delta t F^{n+1}.$$

- ▶ First order in time and implicit.
- ▶ Requires a linear solve at each step.
- ▶ Unconditionally stable in the standard energy sense for the heat equation.

There is no diffusion CFL limit, but accuracy still constrains the time step.

Average the stiffness term between the old and new time levels:

$$M \frac{U^{n+1} - U^n}{\Delta t} + \frac{1}{2} A (U^{n+1} + U^n) = F^{n+1/2}.$$

Higher time accuracy

Second order in time for sufficiently smooth solutions. Like backward Euler, it is implicit.

Balance the spatial and temporal errors

$$\text{error} \lesssim C_s h^p + C_t \Delta t^q.$$

Spatial contribution

Controlled by mesh size, element degree, and solution smoothness.

Temporal contribution

Controlled by the time step and the order of the time integrator.

Reducing one error much below the other adds cost with little benefit.

A worked space–time balance

For a smooth problem with P_1 elements and an L^2 -type spatial error,

Backward Euler

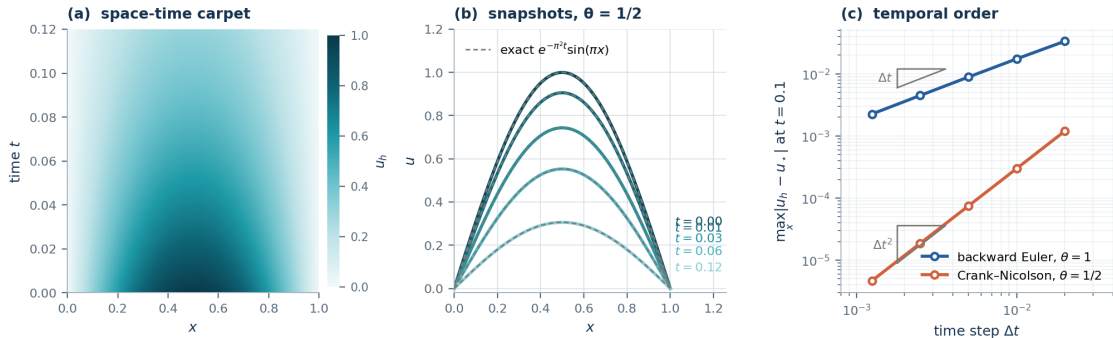
$$h^2 + \Delta t \implies \Delta t \sim h^2.$$

Crank–Nicolson

$$h^2 + \Delta t^2 \implies \Delta t \sim h.$$

These are accuracy balances, not stability restrictions.

What heat diffusion looks like over time



Space-time evolution, solution snapshots, and temporal convergence for the 1D heat equation.

The space–time idea extends to other PDEs

Diffusion

Time stepping must respect dissipative behavior.

Waves and transport

Wave equations have second time derivatives; advection-dominated models may need stabilization.

Reuse the workflow, but recheck stability for the new equation.

Before trusting a simulation

- ▶ Check the weak form and boundary conditions.
- ▶ Check the approximation assumptions and observed convergence.
- ▶ Use error indicators to guide resolution, not as exact errors.
- ▶ For evolution problems, control both space and time errors.

Which result answers which question?

Question	Tool
Does the weak problem have a stable solution?	Lax–Milgram
How good is the discrete approximation?	Galerkin and Céa
How fast does the error decrease?	Interpolation and regularity
Where should the mesh be refined?	A posteriori indicators

From the PDE to a checked solution

Formulate

Write the weak form and choose the trial and test spaces.

Approximate

Choose local polynomial bases and assemble the sparse system.

Verify

Check stability, measure convergence, and estimate the remaining error.